

An Assessment of the Behavior of Compact Thermal Models of Electronic Packages In a Printed Circuit Board Level Environment

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Abstract

This paper outlines research aimed at testing compact thermal models in various board level environments. Results for two generic package types are presented: CPGA and a BGA. A validated fully detailed finite element model and a compact thermal model (CTM) were generated for each. A smeared finite element model of the JEDEC standard thermal test PCB's for each package was built. The boards were modeled assuming both isotropic and anisotropic effective (smeared) thermal conductivities for the PCB. The CTM and fully detailed model (FDM) for each package were "mounted" to its respective test boards. Finite element analysis was performed for each board and package system for a set of five convective boundary sets chosen to test the limits of the models. Runs were also conducted in which heated components were placed with varied proximity to the package model on a larger test board. Comparisons were made of the temperatures predicted by the package FDM / board models and the CTM / board model. The CTM models were found in general to predict junction temperatures to within 5% of the results found using the FDM of the chip packages / board in realistic cooling scenarios. In the cases of high thermal gradients found when edge cooling was the primary means of heat removal, errors rose to a maximum of 9.2%. Despite these higher errors, the average errors for the single component tests were under 3% for both packages. The average error increased to 6-7% with a maximum error of 10.6% for the runs with secondary heat sources. These sources produced high temperature gradients similar to those found in the edge cooling cases. In general, the CTM's only failed to predict the junction temperature of the package to within 5% when high temperature gradients were present in the board.

Key Words: compact thermal models, CPGA, BGA, fully detailed model, printed circuit board, design of experiments

Nomenclature

h	average heat transfer coefficient [W/m ² K]
R	thermal link [°C /W]
T	temperature [°C]
AR	area ratio

Subscripts

t	top
t1, t2	split top areas
ti	top in
to	top out
s	side
bi	bottom in
bo	bottom out
CTM / FDM	compact thermal model of package, fully detailed model of board
FDM / FDM	fully detailed model of both package and board
J	junction
t/b, board	top/bottom of board
edge(s)	edge of board

Introduction

Various procedures for deriving Compact Thermal Models of electronic packages are under development in the thermal community, the most common being the statistical parameter extraction method popularized in Europe [1]. In this methodology, CTM's are derived by exposing experimentally validated finite element models of the package to a set of convective boundary conditions. In this study, the boundary conditions were chosen using the Design of Experiments approach outlined in [2]. In applying these boundary conditions to the finite element model, a set of thermal response data in the form of the junction temperature and heat fluxes through each of the primary sides is generated. A non-linear global optimization tool is used to best fit this data by minimizing a cost function based on the errors found between the CTM and the FDM. The CTM derivation tool ascertains the resistance values and also allows for primary areas to be split into two areas with the area ratio included as an optimized parameter. The resulting simplified resistance network is assumed to be boundary condition independent. This is a highly desirable feature of compact thermal models, for then, the model can be derived exclusive of its board and system level environments.

Whereas there has been much work on the methodology for deriving models, there have been few systematic studies documenting the behavior of a CTM when it interacts with its environment. In general, the boundary condition independence of the CTM is ascertained by comparing its response to a set of boundary conditions to that found from applying that set to the FDM. The validity of representing the

board conduction as an average heat transfer coefficient is not investigated however. In some cases, conduction into a printed circuit board constitutes a significant heat transfer path for an electronic package, especially when cooled by air. For this reason, the assumption that this representation is accurate must be tested at the board level.

Previous Work

Very simple compact models have been tested at the system level and reported in the literature. The most primitive one resistance network (junction to ambient) was placed in a system level environment and exposed to a CFD simulation of the flow in [3]. This was mostly driven by the need to overcome the drastically different length scales between the CFD and chip scale dimensions. The accuracy of this approach was not discussed in comparison to experiment or fully detailed models.

A star-shaped thermal resistance network and a uniform property block representation were tested in a CFD environment and reported in [4]. This used a set of six boundary condition sets to derive the CTM as compared to the exhaustive set suggested in [1] or those found using the design of experiment approach used here. Again this research was performed in part to overcome length scale issues between the device and system. This study compared the CTM's in a system level analysis to a FDM of the entire system and experiments. Reported accuracy was 12-13% from FDM to experiment and 7.5-13% for CTM to FDM comparisons.

A more complex compact thermal model was examined at the system level [5]. Here, the DELPHI approach to develop an optimum network topology was used and tested in a CFD environment. This network was compared to a fully detailed model and a simple two resistance model. The resistance networks were approximated within the system level model by a mixture of high conductivity blocks with surface resistances chosen at the interfaces. The accuracy was reportedly within 5% for the optimized network and 2-20% for the two resistance characterization.

Board conduction independence was addressed in [6]. This study replaced the conduction into the board with effective convective coefficients on the lead surfaces when deriving the compact thermal model. It used a design of experiments approach to obtain a non-redundant set of convective boundary condition sets. The resultant set was far fewer (four) than the extensive set outlined by the DELPHI approach. Arbitrary test cases were then run to test the validity of the compact thermal model.

The work reported in this paper is a continuation on the work reported in [7]. This uses the same technique for testing packages in the board level environment and primarily investigates changes in topology that are not included here but does not address the affect of secondary heat sources on the performance.

Compact Model Generation

Two package types were considered for study. The first was a ceramic pin grid array documented in [2]. As shown in this paper, the package conduction model was experimentally validated. Some slight modifications were made to this

package. First, the original model was validated with the package removed from the socket. This resulted in the inclusion of air between the pins instead of the plastic used in the socket. This was not crucial as the interaction with the board and socket was not investigated within that study. For the purpose of this paper however, the air needed to be replaced by the plastic socket that this package is inserted into when mounted to the board. The second addition was of a copper lid specified by [8]. While these changes did cause the need for re-evaluation of the CTM topology and resistance values and involved approximations, the model is assumed to still be comparably precise to the validated case.

The second package was a ball grid array. A thermal sample, a fully detailed model of the package and board and experimental validation were provided by [9] and [10]. Simpler models were built using this information as a guide. Again, these models were not validated but were instead assumed to be relatively accurate. The cross-section views of each package are shown below in Figures 1 and 2.

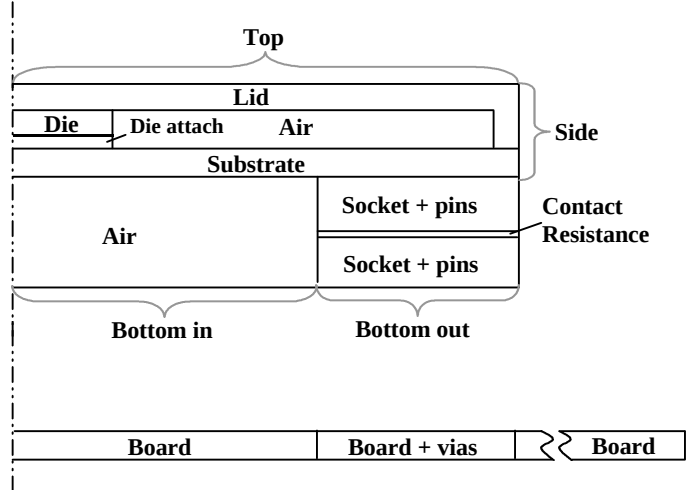


Figure 1: Cross-section view of CPGA package

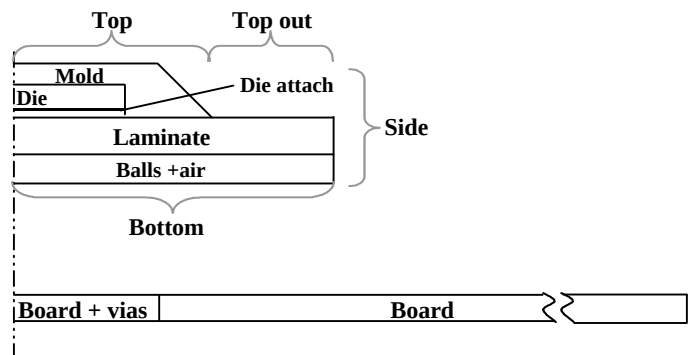


Figure 2: Cross-section view of BGA package

In order to recover this accuracy in the board level models, contact resistances were approximated and placed within the model. The values of these resistances will be "tuned" once experimental validations are run on the thermal test samples.

The compact thermal models were created for these two packages. Convective boundary conditions were placed on the outward facing surfaces noted on Figures 1 and 2 in order to generate the thermal response data needed in the optimization process. These surfaces were divided into four primary lumped surfaces upon which a constant heat transfer coefficient was placed. For the CPGA, these were the top, side, bottom in, and bottom out (leaded) areas. The BGA was divided into a top in, top out, side, and bottom.

The models were exposed to a set of non-redundant boundary conditions as prescribed by the design of experiments approach outlined in [2]. These sets contained eight boundary condition sets covering the maximum and minimum possible heat transfer coefficients for each area. This minimized a cost function specified by [11] while constraints on the maximum allowable error for the prediction of junction temperature (5%) and heat flows (10%) were in place. The top area for the CPGA and the inner top area for the BGA were allowed to be split into two parts by the CTM generation tool and the area ratio was an optimized parameter. Links were removed with each iteration until a reasonable number were attained. This resulted in an eight-link network between the junction and five boundary nodes for each package. The network topologies are shown below in Figures 3a and 3b. The thermal link values are shown below for each in Tables 1 and 2 (##### represents closed links).

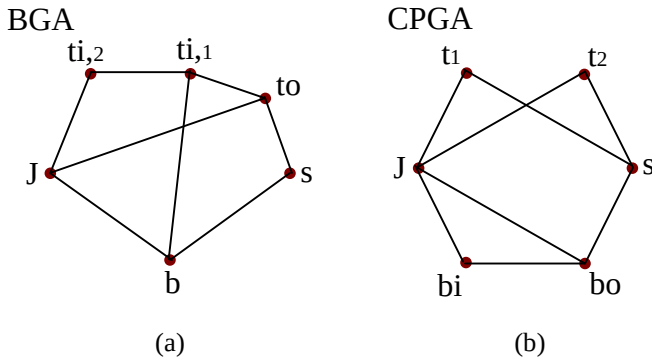


Figure 3: CTM network topologies for the (a) CPGA and (b) BGA chip packages.

CPGA OPTIMUM VALUES OF RESISTORS [K/W]					
Node #	J	t1	t2	s	bi
t1	54.13				
t2	1.27	#####			
s	#####	78.64	0.71		
bi	867.52	#####	#####	#####	
bo	96.89	#####	#####	39.19	1919.70
Top area ratio	AR _t	0.04			

Table 1: Optimized resistance values for CPGA.

BGA OPTIMUM VALUES OF RESISTORS [K/W]					
Node #	J	ti,1	ti,2	to	s
ti,1	34.89				
ti,2	#####	226.59			
to	270.84	#####	17.17		
s	#####	#####	#####	25.20	
b	52.34	#####	18.27	#####	35.05
Top Area Ratio	AR _t	0.24			

Table 2: Optimized resistance values for BGA.

Each network topology is checked against runs using arbitrary boundary condition sets as part of the verification process. This only checks the validity of the model when convective boundary conditions are placed on both the FDM and the CTM of the packages and does not address the ability for the CTM to replace the FDM at the board level. This issue is the focus of the research.

Board Level Modeling

Each package type has a specified JEDEC standard for the thermal test board upon which it should be evaluated in thermal tests [12-14]. The standards most importantly regulate the size of the board and the thickness of the individual trace and dielectric layers. Using these specifications, smearing can be utilized to approximate the in-plane and thru-plane conductivities for a 1S0P and a 1S2P thermal test board. In order to fully test the potential influence of board conductivity on the accuracy of the CTM, several isotropic board conductivities were also chosen. These covered three orders of magnitude ($k = 2, 10, 100$ W/m-K).

Both boards contained areas where vias were present to aid in transporting signals and heat to different layers within the board. Fortunately, in both cases these areas were well defined. These areas were smeared to find an effective thermal conductivity in the same fashion as was employed for the pins in the CPGA model.

The validated FDM and derived CTM for each chip package were placed onto each of these five boards. The cases with fully detailed models of both the chip package and board (FDM / FDM) were assumed to be the “true” cases. When the CTM of a package was placed on a FDM of the board (CTM / FDM), it was referred to as the “numerical” case. Figures 4 and 5 on the following page show each case for the CPGA package.

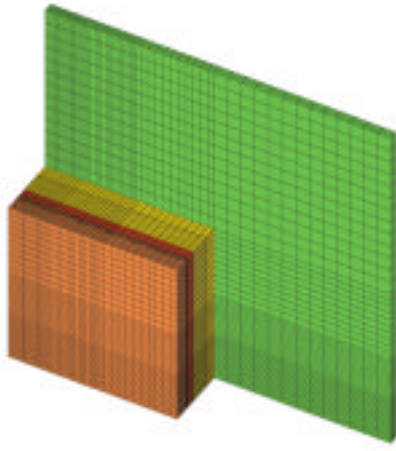


Figure 4: Fully detailed model of board and CPGA package (FDM / FDM).

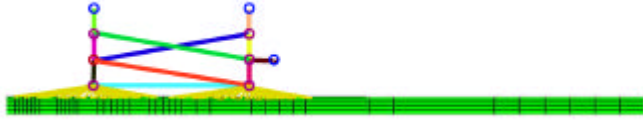


Figure 5: Compact thermal model of CPGA package on fully detailed model of board (CTM / FDM).

A set of test cases was developed to adequately assess the CTM / FDM case. Each case was chosen to test a different extreme of the board level model. Case A tested the response when the top and side (ie, the surfaces of the package) were the primary heat paths. Case B was chosen as an arbitrary and more realistic case. Case C used the top and bottom surfaces of the board as the primary heat paths. Cases D and E were edge-cooling scenarios. Case D was a non-symmetric case where convection was applied to one of the four edges of the PCB. Test Case E had heat removal primarily through all four edges. These are listed in Table 3.

BC Set	h_t	h_s	$h_{t/b, \text{board}}$	$h_{\text{edge(s)}}$
A	1000	1000	5	0
B	800	250	50	0
C	5	5	1000	0
D	5	5	0	1000
E	5	5	0	1000

Table 3: Boundary condition sets for board level models

Interactions when other heat sources were present on the board were also of interest. For this, the PCB models were extended so that a secondary heat source could be placed in proximity to the package. This heat source was set to dissipate the same amount of heat as the die within the package and was sized at 35 [mm] x 35 [mm]. Heat sources were placed at (three) varying distances from the package and the junction temperature was evaluated for the FDM / FDM

and CTM / FDM cases as before. The board was lengthened and laid out as shown below in Figure 7.

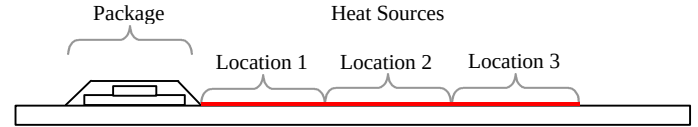


Figure 6: Layout for dual-component runs

These heat sources provided a thermal gradient that differs from the single component cases. Each of the three source locations was evaluated for the three isotropic board cases. A constant heat transfer coefficient of 15 [W/m²-K] was placed on all surfaces.

Results

For the single component runs, the CTM / FDM models were generally found to predict the junction temperature to within 3% of the “true” model for boundary condition sets A, B, and C. For the edge cooling cases however, the errors were usually much higher; some as high as 9.2%. This was caused by the inability of the CTM to properly handle the large temperature gradients within the board near the package. These temperature changes and the resulting heat flow pattern cannot be accurately represented when the entire area is modeled as one or two nodes.

The network of open thermal links leading up to these nodes also allow for redistribution of heat on the surface of the board. This basically changes the conditions that the bottom surface of the package faces. This can be seen in Figure 7 by the isothermal area found in the package area for the CTM / FDM case. The gradients found in the remainder of the board however are comparable. Since the bottom node of the package must approximate this area as one temperature, this may not be a source of error in this process. If these connecting links can be replaced by resistances that allow heat flow in one direction only, it will not allow for the lateral distribution of heat on this surface and hence will retain the board gradients.

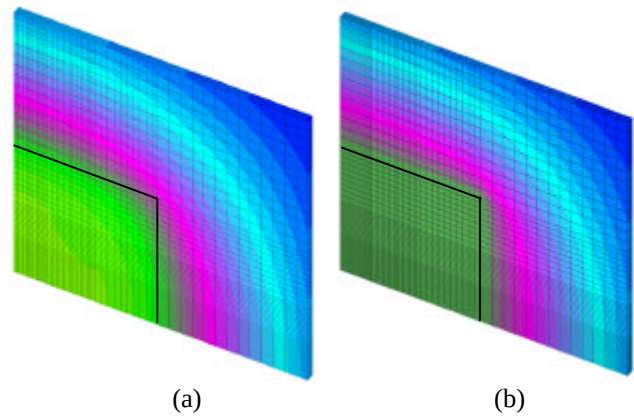


Figure 7: Board temperature contours for CPGA with isotropic board conductivity ($k = 10$ W/m-K) subjected to Boundary Condition Set E. Both cases (a) FDM / FDM and (b) CTM / FDM are shown.

The relative errors between the numerical and the “true” values are shown in Figures 8 and 9. The dashed lines on each represent the maximum error allowed during the optimization process. The area below this line is referred to as the fidelity region. Any error within this region is inherent to the optimization process and is acceptable.

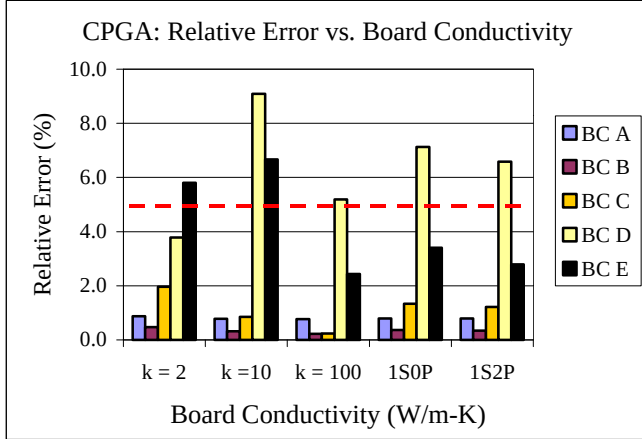


Figure 8: Relative error for CPGA package for varied board conductivities and boundary condition set.

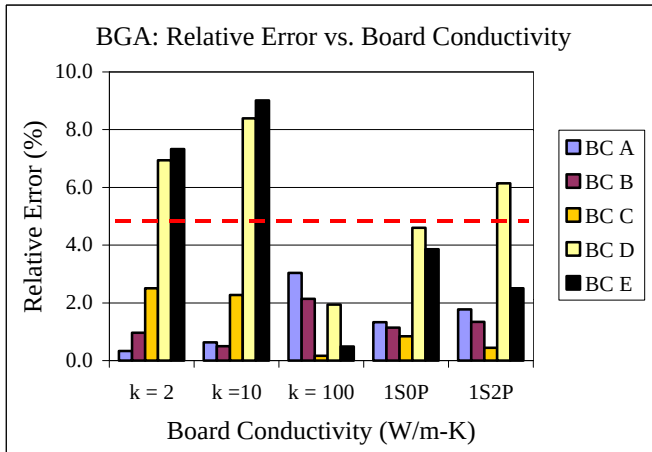


Figure 9: Relative error for BGA package for varied board conductivities and boundary condition set.

For the CPGA, it can be readily seen that the errors for the first three boundary condition sets (A, B, and C) that the errors are within the fidelity region of 5%. For the final two boundary condition sets however the error increases and in fact exceeds the error that is deemed inherent to the optimization process. These additional errors are attributed to the board gradients caused by these edge-cooling scenarios. Even for these extreme cases however, the errors are within 10%. The average error for all 25 single component, CPGA trials was 2.6%.

The BGA had similar results. Again, the first three boundary condition sets have very reasonable accuracy. The edge cooling cases (D and E) strayed outside of the fidelity region for some cases. The errors were within 10% and the average error is up slightly to 2.8%.

The BGA package showed greater sensitivity to board conductivity than the CPGA package. This is likely due to the fact that this package is more thermally coupled to the board. This coupling caused more variation in the error with changes in board conductivity.

It can be clearly seen for both cases that the errors for the moderate conductivity case of 10 W/m-K are the highest. This is caused by the high temperature gradients. For the higher conductivity cases found in the 1S0P, 1S2P, and the isotropic board with a value of 100 W/m-K, these gradients are smoothed. For the lowest conductivity case, the board becomes a minor heat flow path.

Results With Secondary Heat Source

The results were similar for the cases using a secondary source. These can be seen below in Figures 10 and 11. These cases most closely compare with the non-symmetric edge cooling case for the single component runs (Case D). Figure 6 shown previously displays the layout for the three locations.

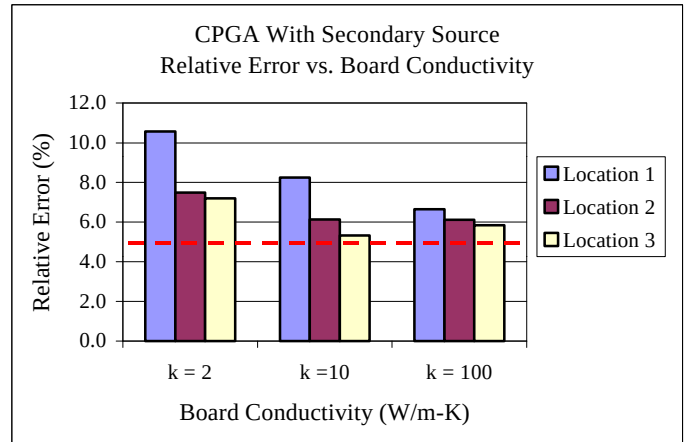


Figure 10: Relative error for CPGA package with secondary source for varied isotropic board conductivities and locations.

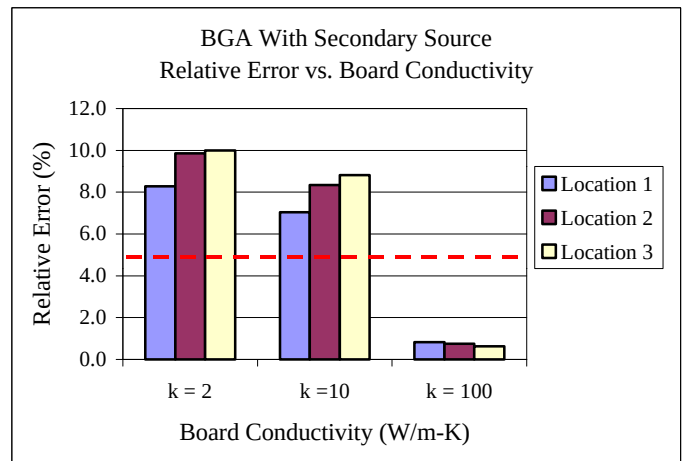


Figure 11: Relative error for BGA package with secondary source for varied isotropic board conductivities and locations.

Only one set (BGA, $k=100$ W/m-K) of data falls within the fidelity region. The rest are comparable to the cases found in Case D for the single source runs.

The presence of board gradients and the difficulty in the prediction of these gradients by the CTM are best shown in the board gradients for the FDM / FDM cases as compared to those of the CTM / FDM cases. A sampling of these for the BGA package are shown in Figures 12-16.

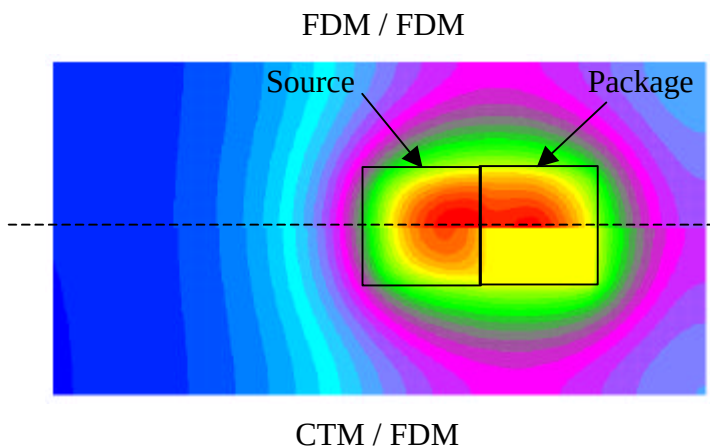


Figure 12: Board Contours for BGA with secondary source in Position 1 on $k = 10$ W/m-K board.

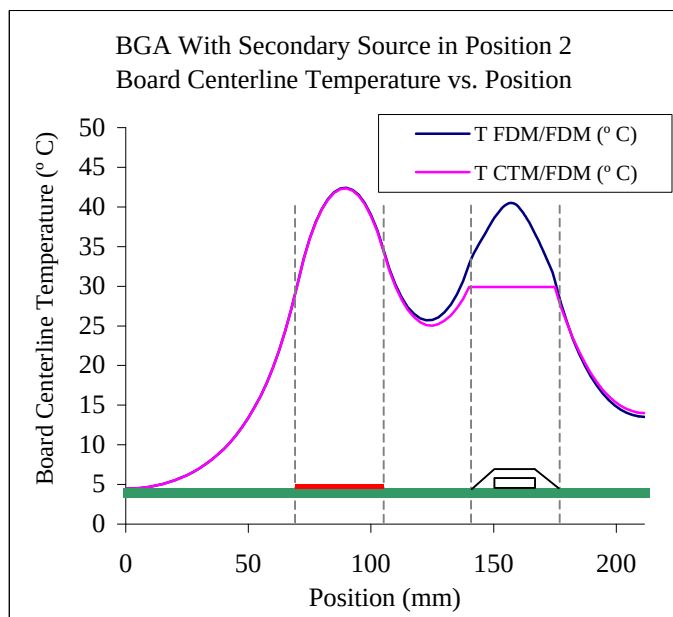


Figure 13: Board centerline temperature vs. location on the board for the BGA with secondary source in position two.

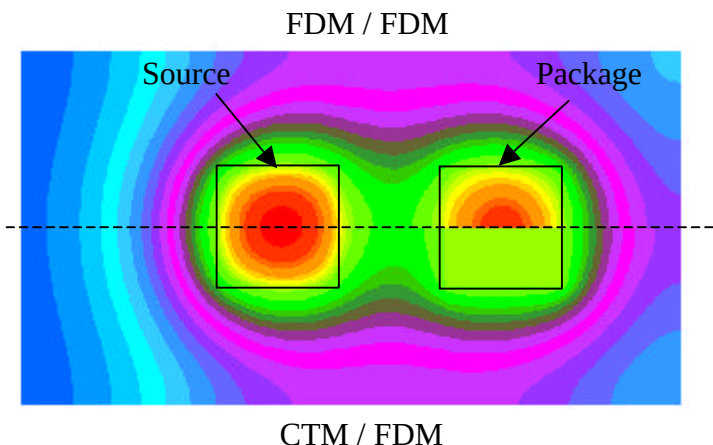


Figure 14: Board contours for the BGA with secondary source in Position 2 on $k = 10$ W/m-K board.

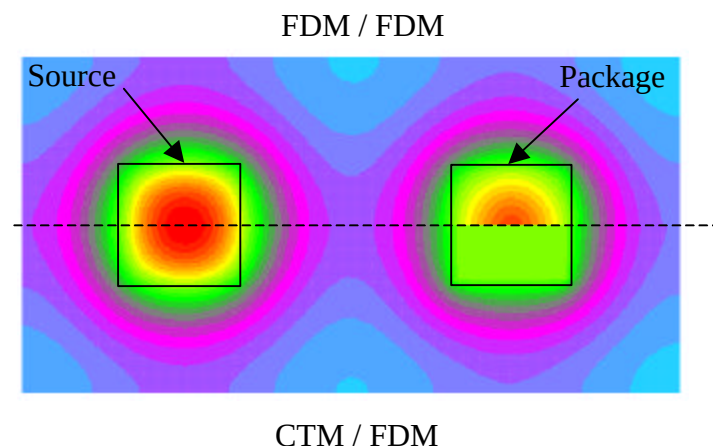


Figure 15: Board Contours for BGA with secondary source in Position 3 on $k = 10$ W/m-K board.

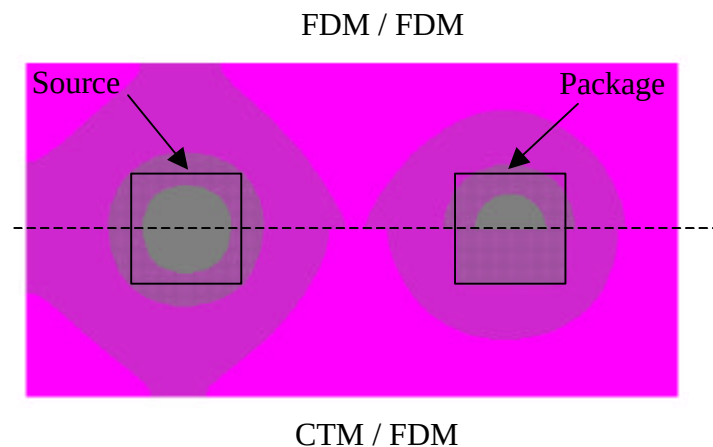


Figure 16: Board Contours for BGA with secondary source in Position 3 on $k = 100$ W/m-K board (same temperature scale as Figure 15)

Figures 12, 14, and 15 show how the contours change as the heat source moves away from the package for the moderate conductivity value of 10 W/m-K. For each of the contour plots, the area beneath the package is forced by the CTM attachment process to be isothermal for the CTM / FDM case as discussed previously. This is best seen by the results shown in Figure 13 for each case. The board centerline temperatures are almost perfectly predicted by the CTM / FDM model except in the region where the package is attached to the board. The temperatures within the board are found relatively accurately outside of this region for all of the runs. As would be expected, the further the source is placed from the package, the less error is seen in the contours.

Figure 16 shows the effect of the highest conductivity on the temperature gradients. This compares to Figure 15 as both are plotted over the same temperature range. The approximation made by the CTM / FDM models of the bottom surface being isothermal is not as poor for the high conductivity board (Fig. 16) as the moderate value (Fig. 15).

Summary and Conclusions

Fully detailed package models were developed for two package types: a BGA and a CPGA. Each was used to develop a compact thermal model for the package type. The FDM and the CTM of the package were each mounted to a fully detailed model of the JEDEC thermal test board for its package type. The thermal conductivity of the test board was varied to represent a smeared model of the 1S0P and 1S2P JEDEC boards. Isotropic boards were also used to evaluate the dependence of accuracy on board conductivity. These were set to cover three orders of magnitude. Each of the two setups (CTM / FDM and FDM / FDM) was run through five different boundary condition sets to test extreme cases the CTM may experience. Secondary sources were added with varied proximity to the package on an elongated. These were tested for all three isotropic board conductivities.

The major observations made were the following:

1. The average error was about 3% (well within the fidelity limit of 5% for the optimization process) for the cases without a secondary source. When the secondary heat source was in place the errors rose to 6-7%.
2. Higher errors were generally found when high temperature gradients resulted in the board (boundary condition sets D and E and the secondary source cases)
3. The moderate conductivity boards generally produced the largest errors. The higher conductivity boards showed the least error due to the smoothing of the temperature gradients. The lowest conductivity boards decreased the heat flow decreasing the importance of the board model.
4. The BGA showed more variation in the error because of the increased thermal coupling between the package and board for this package family.
5. The greatest discrepancy in the models actually occurs on the surface of the board located beneath the package. Due to the need to represent that surface as one node, this area is approximated as isothermal by the solver. This does not have a deleterious affect on the prediction package level temperatures and heat flows.

Acknowledgements

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